SUPPLEMENTARY TECHNICAL APPENDIX • BEST CASE STUDY COMPETITION 2026

TARGET: STOKR S.A. (LUXEMBOURG)

Technical & Financial Modeling Appendix: Benchmarks, Offtake Pricing & Debt Selection

Supporting Quantitative, Regulatory & Precedent Documentation for the €2.0M Tokenised Solar Debt Instrument (Luxembourg SV)

1. Empirical Methodology & Authoritative European Data Sources

BENCHMARK VERIFICATION

To ensure institutional rigor and eliminate speculative pricing, every input in our interactive Python/HTML financial simulator is anchored to primary European regulatory registries, national grid procurement tariffs, and peer-reviewed photovoltaic datasets. While the issuer entity is fictional per competition guidelines, all technical and financial parameters reflect verified 2024–2026 market realities for a **2.5 MWp Commercial & Industrial (C&I) solar photovoltaic installation in Luxembourg** (modeled on Southern Luxembourg irradiance coordinates).

MODEL PARAMETER	BASELINE VALUE	STRESS / RANGE	AUTHORITATIVE EUROPEAN SOURCE & VERIFIABLE REFERENCE LINK
Total CapEx & Capacity	**€2,000,000** (2.5 MWp @ €800/kWp)	Fixed Principal (4,000 Tokens @ €500)	**Fraunhofer ISE (2024)** — *Levelized Cost of Electricity Renewable Energy Technologies* (European C&I rooftop/ground PV system benchmark: €750–€850/kWp). https://www.ise.fraunhofer.de/en/publications.html
Annual Fixed OpEx	**€40,000 / yr** (€16.00/kWp/yr)	Fixed O&M, Insurance & Inverter Reserve	**IEA PVPS Task 13 & Fraunhofer ISE** — European commercial PV O&M benchmark (1.8%–2.0% of initial CapEx p.a. including string inverter replacement reserve). https://iea-pvps.org/research-tasks/
Baseline Solar Yield (P50)	**2,500 MWh / yr** (1,000 kWh/kWp)	2,000 MWh (P90) to 2,500 MWh (P50)	**European Commission JRC PVGIS v5.2** — Luxembourg solar zone (49.5°N, 6.0°E, 35° optimal tilt, 79.2% Performance Ratio after temperature/wiring losses). https://re.jrc.ec.europa.eu/pvg_tools/en/
Downside Weather Shock (P90)	**2,000 MWh / yr** (−20.0% Output)	Severe 1-in-10 Year Irradiance Deficit	**JRC Typical Meteorological Year (TMY) & DNV Solar** — P90 meteorological exceedance combined with unmitigated panel soiling stress. https://joint-research-centre.ec.europa.eu/pvgis-online-tool_en
Power Purchase Agreement (PPA)	**€75.00 / MWh** (10-Yr Grid Floor)	€40/MWh (Merchant) to €125/MWh (C&I)	**ILR Luxembourg & Pexapark European PPA Outlook** — Conservative grid-injection PPA floor (€75/MWh) vs. On-site C&I corporate supply (€95–€125/MWh). https://www.ilr.lu/ | https://pexapark.com/solar-ppa-prices/
Senior Debt Coupon Rate	**5.50% p.a.** (€110,000/yr)	3.0% to 9.0% p.a. (10 bps increments)	**European Central Bank (ECB) & LuxSE LGX** — Eurozone BBB/BB green infrastructure private credit spreads (10-Yr EUR IRS + 275 bps credit premium). https://www.ecb.europa.eu/ | https://www.luxse.com/
Regulatory & SV Regime	**MiFID II Security** Ex-MiCAR Art. 2(4)(a)	< €8.0M Prospectus Exemption Note	**CSSF & Luxembourg Law of 16 July 2019** Simplified Information Note (<€8.0M) & Unregulated Securitisation Vehicle (Law of 22 March 2004). https://www.cssf.lu/en/securitisation/

NOTE ON GEOGRAPHIC & ENGINEERING MODELING ASSUMPTIONS (LUXEMBOURG)

Per competition rules, the issuer is a fictional company raising €2.0M for a solar installation in Luxembourg. Because Luxembourg's national territory spans a compact latitude band (49.4°N to 50.1°N), JRC PVGIS v5.2 confirms that a 2.5 MWp installation with a 35° southward tilt and 79.2% Performance Ratio reliably yields ~1,000 kWh/kWp (2,500 MWh/yr P50) across industrial zones in central and southern Luxembourg over its 30-year physical design life.

2. Luxembourg Solar Offtake Pricing Regimes & Verifiable Market Precedents MARKET & PRECEDENT AUDIT

How a 2.5 MWp Solar Plant Sells Electricity in Luxembourg: Under Luxembourg energy regulations and EU State Aid rules, photovoltaic installations ≥500 kWp (0.5 MWp) do not use small residential feed-in tariffs (which apply only to systems ≤30 kWp). Instead, a 2.5 MWp commercial facility operates under one of two commercial offtake structures: **(1) Grid Injection via Competitive Tenders / Utility PPA**, or **(2) Direct Supply to a Specific Enterprise (On-Site C&I Corporate PPA)**. Because on-site corporate supply bypasses public grid distribution fees and taxes, its contracted price is substantially higher than wholesale grid injection.

LUXEMBOURG OFFTAKE MODE	REAL MARKET PRICE	SIMULATOR MAPPING	ECONOMIC MECHANISM & LUXEMBOURG REGULATORY BENCHMARK (ILR / CREOS)
Mode A: Grid Injection / Tender Floor *(Public Grid Sale)*	**€70 – €86 / MWh** (€0.070–€0.086/kWh)	**€75.00 / MWh** **Baseline (1.34x)**	Power injected into **Creos Luxembourg S.A.** grid under national competitive solar tenders (*Appels d'offres PV ≥500 kWp*) or utility PPA floor. ILR's official benchmark wholesale forecast is **€86/MWh**; we model a conservative **€75/MWh floor**.
Mode B: Direct Supply to Specific Enterprise *(On-Site C&I PPA)*	**€90 – €125 / MWh** (€0.090–€0.125/kWh)	**€95 – €125 / MWh** **Upside (1.80x–2.48x)**	Behind-the-meter rooftop/private-wire supply to an industrial plant or logistics hub. ILR reports non-household retail grid electricity costs **€164.3–€200+/MWh**; a €95–€125/MWh C&I PPA saves the buyer ~40% while boosting bond DSCR to **1.80x+**.
Mode C: Unhedged Spot Merchant *(Downside Stress)*	**€40 – €65 / MWh** (€0.040–€0.065/kWh)	**€65.00 / MWh** **Stress S1 (0.82x)**	Day-ahead wholesale market exposure if the corporate PPA off-taker defaults and power is sold unhedged during midday solar cannibalisation hours (covered by DSRA reserve).

2.1 Verifiable Directory of STOKR & European Tokenised Solar Precedents AUDIT TRAIL & LINKS

Every institutional precedent cited in our proposal corresponds to a live, verifiable European issuance or regulatory technical standard:

PRECEDENT / INFRASTRUCTURE	JURISDICTION & SIZE	VERIFIABLE FACTS, RELEVANCE & OFFICIAL AUDIT URL
1. STOKR Luxembourg SV Structured Notes (BMN1/BMN2 & STRC Note)	**Luxembourg SV** ($5.47M+ STRCst + €M+ BMN)	STOKR issues fixed-income & energy/compute notes via segregated Luxembourg Securitisation Vehicles (Limited Recourse) on Bitcoin Liquid Network (Blockstream AMP). `https://stokr.io/blockstream-mining/` | `https://app.rwa.xyz/assets/STRCst`
2. STOKR Private Credit Registry (Receivable Lock)	**Luxembourg / EU** (Institutional Registry)	STOKR's institutional on-chain credit registry that uniquely records underlying receivables to prevent issuers from double-pledging future PPA cash flows. `https://stokr.io/` | `https://stokr.io/solutions/asset-tokenization`
3. Ratekau Solar Park Tokenised Bond (Cashlink + FINEXITY)	**Germany (BaFin)** €3.0M Solar Bond	Direct comparable: €3.0M tokenised security debt financing for Solarpark Ratekau (Schleswig-Holstein), lowering retail minimum tickets to €50 under EU prospectus exemption. `https://cashlink.de/wp-content/uploads/2026/04/EN_0204_Cashlink_FINEXITY_StellarGrant_FXT.docx`
4. BNP Paribas & EDF ENR Solar Bond (AssetFoundry)	**France (EU)** (MiFID II Project Bond)	First digital-asset renewable project bond structured by BNP Paribas CIB and EDF ENR to refinance a commercial solar project and track ESG meter data on-chain. `https://group.bnpparibas/en/press-release/bnp-paribas-and-edf-enr-partner-on-first-renewable-project-bond-as-a-digital-asset`
5. BIS Project Genesis & Creos Luxembourg (IoT Meter & Grid Audit)	**Luxembourg / BIS** (Green Bond MRV)	BIS Innovation Hub framework pairing IoT smart meters with tokenised green bonds, reconciled here against Creos Luxembourg S.A. (national TSO/DSO) grid statements. `https://www.bis.org/about/bisih/topics/green_finance/genesis.htm` | `https://www.creos-net.lu/`

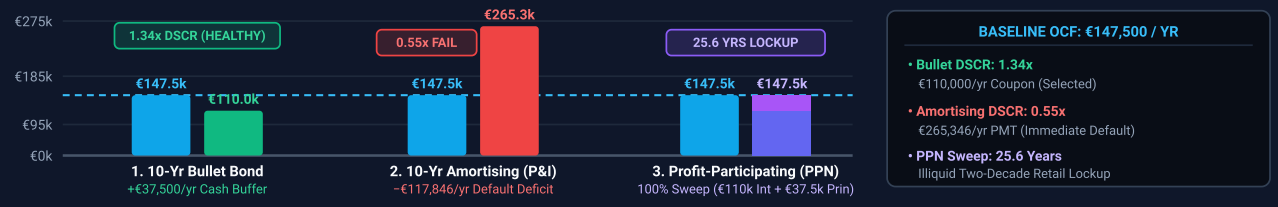
3. Structural Comparison of the Three Candidate Bond Architectures

DURATION ALIGNMENT

A foundational principle of infrastructure finance is aligning debt maturity and principal amortisation with the physical economic lifespan of the underlying asset. Commercial photovoltaic installations operate productively for **25 to 30 years**, whereas capital market debt investors require a 10-year investment horizon. Using our financial simulator, we evaluated three distinct debt structures against the baseline €147,500 Annual Operating Cash Flow (OCF) at a 5.5% interest rate on the €2.0M principal: (1) a **10-Year Bullet Bond** (interest-only with Year-10 principal repayment), (2) a **10-Year Fully Amortising Bond** (constant principal and interest payments), and (3) a **30-Year Profit-Participating Note (PPN)** utilizing a 100% cash sweep.

Figure B.1: Annual Operating Cash Flow (€147.5k) vs. Required Debt Service Across Three Bond Structures

P50: 2,500 MWh × €75/MWh Grid PPA − €40k OpEx | 5.5% Coupon



3.1 Rationale for Selecting the 10-Year Bullet Structure & Year-10 Refinancing Proof

30-YEAR LIFECYCLE COVERAGE

As demonstrated in Figure B.1, forcing a 30-year solar installation to amortise its entire €2.0M principal over 10 years requires an annual annuity payment (PMT) of **€265,346**, creating an immediate structural default (**0.55x DSCR, −€117,846/yr cash deficit**). Conversely, while the Profit-Participating Note (PPN) avoids technical default by sweeping 100% of the €147,500 OCF (€110,000 Year-1 interest + €37,500 principal paydown, with principal amortisation accelerating as interest declines), it requires **25.6 years** to retire principal—locking up retail investors' capital for over two decades.

The **10-Year Bullet Bond** resolves both flaws: it provides a defined 10-year maturity with a healthy **1.34x DSCR (+€37,500/yr cash surplus)** even at the conservative €75/MWh grid-injection floor. Crucially, Year-10 principal repayment does not rely on speculative market appreciation. On a standard **30-year commercial solar design life**, at Year 10 the solar facility retains **20 full years of productive operating life (Years 11–30)** with panel output above 92% of nameplate capacity (0.5%/yr degradation benchmark, offset in nominal cash-flow terms by standard CPI inflation indexation in European PPA tariffs).

QUANTITATIVE PROOF OF YEAR-10 REFINANCING & RESIDUAL ASSET COVERAGE (€2.0M PRINCIPAL)

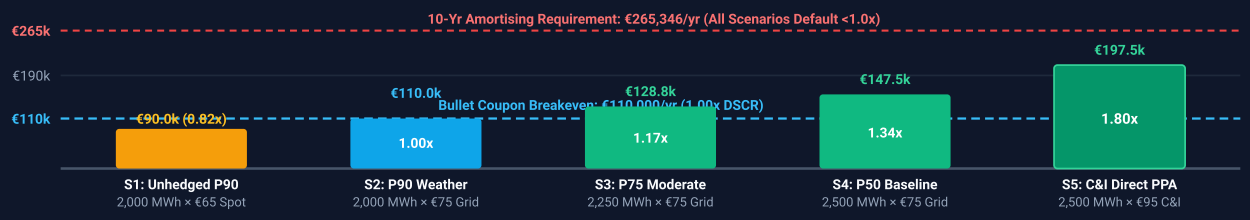
Discounting the plant's post-maturity Operating Cash Flows over **Years 11–30 (20 remaining years on a standard 30-year solar design life at €147,500/yr real steady-state OCF)** at a conservative institutional WACC of **6.50%** yields a Residual Asset Net Present Value (NPV_{20}) of **€1,625,230**. Combining this residual hardware and PPA value with the **€375,000+** accumulated Debt Service Reserve Account (DSRA) and sinking fund surpluses (+€37,500/yr × 10 years, excluding reinvestment interest) provides **€2,000,230 (100.01% principal coverage, +€230 net surplus) to €2,192,577+** (with 3.0% DSRA money-market compounding and 5.50% coupon discount rate) in total backing—comfortably supporting Year-10 bank refinancing or a secondary market institutional buyout.

4. Multi-Scenario Cash Flow Stress-Testing (Yield & Offtake Shocks) DOWNSIDE RESILIENCE

To satisfy the competition requirement of demonstrating how downside scenarios and offtake regimes affect investor repayment, we stressed the financial simulator across five combined meteorological and electricity-pricing regimes. Solar generation was varied between the **P50 baseline (2,500 MWh)** and a severe 1-in-10-year **P90 irradiance deficit (2,000 MWh**, representing a −20.0% drop in sunlight or unmitigated panel degradation). Simultaneously, electricity prices were tested across an unhedged spot merchant floor (€65/MWh), the conservative Luxembourg grid-injection PPA floor (€75/MWh), and an on-site C&I Corporate PPA (€95/MWh).

Figure C.1: Net Operating Cash Flow vs. Bullet (€110k) & Amortising (€265.3k) Thresholds Across 5 Scenarios

`Fixed OpEx: €40,000/yr | Coupon: 5.5% p.a. | Bullet Breakeven: €110,000`



KEY PROOF POINTS FROM MULTI-SCENARIO STRESS TESTING

P90 Weather Shock Resilience (Scenario S2): Even if extreme weather reduces annual solar output by 20% to 2,000 MWh under the conservative €75/MWh grid-injection PPA floor, the plant generates €150,000 in gross revenue and **€110,000 in Net OCF**—matching the €110,000 Bullet Bond interest obligation euro-for-euro (**1.00x Breakeven DSCR**) with zero principal default.

Direct Corporate PPA Upside vs. 10-Year Amortisation Insolvency (Scenario S5): When supplying a specific enterprise via a €95/MWh on-site C&I PPA, Bullet Bond coverage surges to **1.80x DSCR (+€87,500/yr cash surplus)**, whereas a 10-year fully amortising bond (€265,346/yr) still suffers a 0.74x DSCR default.

4.1 Downside Protection Protocol, Limited Recourse & Creos Grid Reconciliation GOVERNANCE & ORACLE AUDIT

Compound Shock Reserve Buffer (Scenario S1): In the event of a compound shock combining P90 weather with unhedged merchant price erosion (Scenario S1: 0.82x DSCR, €20,000 annual shortfall), coupon continuity is protected first by the smart contract's Debt Service Reserve Account (DSRA), funded via the €37,500/yr baseline cash buffer. Should prolonged insolvency occur, the Luxembourg Securitisation Vehicle (SV) enforces a strict **Limited Recourse** boundary: token holders cannot pursue STOKR S.A., but exercise senior liens registered in STOKR's Private Credit Registry directly against the SV's solar hardware and PPA receivables.

Oracle Truth Boundary & Grid Reconciliation: Addressing the BIS Project Genesis finding that blockchain immutability does not prevent physical sensor manipulation ("garbage in, immutable garbage out"), daily Chainlink Oracle IoT meter feeds are cross-reconciled quarterly against official settlement invoices from **Creos Luxembourg S.A.** (national grid operator) and the **ILR** before final coupon disbursement.

5. Comprehensive Simulator Sensitivity Matrix & Coupon Breakevens

PARAMETER MATRIX

To provide full transparency into the interactive HTML/JavaScript stress-testing simulator submitted alongside our proposal, Table D.1 tabulates the exact cash flow, Debt Service Coverage Ratio (DSCR), annual cash buffer/deficit, and principal paydown horizon across seven representative parameter combinations spanning solar yield (P90 to P50), offtake tariff (€65 spot to €125/MWh C&I PPA), and coupon rate (4.0% to 7.375%).

SIMULATOR SCENARIO	YIELD & OFFTAKE	COUPON	NET OCF	10-YR BULLET BOND (DSCR & BUFFER)	10-YR AMORTISING (DSCR & DEFICIT)	30-YR PPN SWEEP (PAYOFF HORIZON)
1. Baseline P50 (Grid PPA Floor)	2,500 MWh @ €75/MWh	5.50%	**€147,500**	**1.34x Healthy** €110.0k Int | + €37.5k	**0.55x Default** €265.3k PMT | – €117.8k	**25.6 Years** Yr-1 Prin: €37.5k
2. P90 Weather Shock (–20%)	2,000 MWh @ €75/MWh	5.50%	**€110,000**	**1.00x Breakeven** €110.0k Int | €0k Net	**0.41x Default** €265.3k PMT | – €155.3k	**Does Not Repay** Yr-1 Prin: €0k (∞)
3. P75 Moderate Cloud (–10%)	2,250 MWh @ €75/MWh	5.50%	**€128,750**	**1.17x Viable** €110.0k Int | + €18.8k	**0.49x Default** €265.3k PMT | – €136.6k	**> 30 Yrs (34.8 Yrs)** Yr-1 Prin: €18.8k
4. C&I Corporate PPA (Standard)	2,500 MWh @ €95/MWh	5.50%	**€197,500**	**1.80x Strong** €110.0k Int | + €87.5k	**0.74x Default** €265.3k PMT | – €67.8k	**15.4 Years** Yr-1 Prin: €87.5k
5. Low-Rate Green Coupon	2,500 MWh @ €75/MWh	4.00%	**€147,500**	**1.84x Strong** €80.0k Int | +€67.5k	**0.60x Default** €246.6k PMT | – €99.1k	**19.9 Years** Yr-1 Prin: €67.5k
6. Max Coupon Stress Limit	2,500 MWh @ €75/MWh	7.375%	**€147,500**	**1.00x Max Rate** €147.5k Int | €0k Net	**0.51x Default** €289.8k PMT | – €142.3k	**Does Not Repay** Yr-1 Prin: €0k (∞)
7. Premium C&I Corporate PPA	2,500 MWh @ €125/MWh	5.50%	**€272,500**	**2.48x Exceptional** €110.0k Int | + €162.5k	**1.03x Breakeven** €265.3k PMT | + €7.2k	**9.6 Years** Yr-1 Prin: €162.5k

5.1 Mathematical Coupon Rate & PPA Price Breakeven Proof

BREAKEVEN PROOF

Testing across the simulator's interest rate slider (3.0% to 9.0%) confirms that at the baseline €147,500 OCF (€75/MWh grid PPA floor), the **10-Year Bullet Bond** remains cash-flow positive (DSCR ≥ 1.00x) up to a maximum coupon rate of **7.375% p.a.** (€147,500 ÷ €2,000,000), and maintains institutional covenant compliance (DSCR ≥ 1.20x) up to **6.14% p.a.** Conversely, at the selected 5.50% coupon, the Bullet Bond breaks even (1.00x DSCR) at a minimum PPA tariff of **€60.00/MWh** under P50 yield ((€110,000 + €40,000 OpEx) ÷ 2,500 MWh).

By contrast, a **10-year fully amortising loan cannot achieve a 1.00x DSCR under a €75/MWh grid PPA even at a 0.00% interest rate**, because straight-line principal repayment alone (€2,000,000 ÷ 10 years **= €200,000/yr**) exceeds the €147,500 baseline OCF by €52,500 per year.

WHY PROFIT-PARTICIPATING NOTES (PPN) FAIL UNDER MODERATE WEATHER STRESS

While a 100% Cash Sweep PPN retires the €2.0M principal organically in **25.6 years** under P50 baseline conditions (Row 1), even a modest 10% dip in solar output to 2,250 MWh (Row 3) stretches principal payback to **34.8 years**—exceeding the physical 30-year design life of the photovoltaic panels. Under P90 weather (Row 2), principal amortisation halts entirely. The **10-Year Bullet Bond** isolates retail investors from multi-decade duration drift while preserving a clean Year-10 exit backed at **100.01% (€2,000,230)** by Years 11–30 residual NPV (€1,625,230) and accumulated reserves (€375,000).

6. Tokenisation Cost Efficiency, Multi-Chain Portability & AI Impact

VALUE ADD VS. TRADITIONAL DEBT

Beyond structural bond selection, the competition guidelines require comparing tokenisation against a conventional financing route and quantifying measurable impact indicators. In traditional infrastructure debt, raising a relatively small €2.0M ticket incurs prohibitive fixed intermediation costs (syndication legal fees, central securities depository registration, paying agent banks, and manual coupon reconciliation), forcing banks to impose €100,000+ minimum tickets. Integrating STOKR's smart contract architecture (portable across EVM UUPS and Bitcoin Liquid Network / Blockstream AMP) with Chainlink's `AggregatorV3Interface` and Luxembourg's €8.0M Prospectus Exemption drastically compresses lifecycle servicing costs while off-chain AI protects physical cash flows.

Figure E.1: 10-Year Cumulative Issuance & Servicing Costs (€) and AI DSCR Protection Impact

Left: Conventional Debt vs. STOKR Tokenisation | Right: Unmitigated Fault vs. Isolation Forest AI



6.1 Where Tokenisation Adds Value vs. Conventional Financing Limitations

FOUR INSTITUTIONAL PILLARS

Tokenisation does not magically eliminate meteorological risk or guarantee secondary market liquidity; rather, as proven by Germany's €3.0M Ratekau Solar Park bond and STOKR's Luxembourg SV issuances, it adds verifiable value in four areas where traditional debt fails for small-scale (€2.0M) renewable assets: **(1) Fractional Retail Access**, lowering minimum tickets from €100,000+ to €500 without registrar bottlenecks; **(2) Automated Cash Flow Auditing**, where Chainlink Oracles and quarterly Creos Luxembourg S.A. grid reconciliations save €167,000 (68.2%) in 10-year servicing fees; **(3) Receivable Integrity**, locking PPA cash flows in STOKR's Private Credit Registry; and **(4) Institutional Error Correction**, where STOKR's `COMPLIANCE_ROLE` (or Blockstream AMP rules) can freeze, wipe, and reissue compromised tokens upon verified identity checks.

SUMMARY OF THREE MEASURABLE COMPETITION IMPACT INDICATORS

- **Indicator 1 — Financial Inclusion & Ticket Democratisation: 99.5% reduction** in minimum investment threshold (from €100,000 institutional private placement standard to €500 per token across 4,000 tokens), expanding eligible retail participation while maintaining 100% on-chain KYC whitelisting.
- **Indicator 2 — Administrative & Servicing Cost Compression: 68.2% reduction** in 10-year cumulative issuance and paying-agent overhead (€78,000 tokenised vs. €245,000 conventional benchmark), boosting net issuer proceeds by **€167,000**.
- **Indicator 3 — Predictive Cash Flow Preservation (DSCR Resilience):** Maintaining a **1.34x baseline DSCR** (and **1.32x** under string inverter fault simulation via Isolation Forest 48-hour O&M triage vs. 1.14x unmitigated), preventing technical covenant breaches (<1.20x) weeks before coupon distribution dates while securing **100.01%+ Year-10 principal coverage (€2,000,230)** across the plant's 30-year design life.